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ENGINEERING



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Trust Scan QR: Intelligent Java-Based System for Detecting Malicious QR Codes with Risk Scoring and URL Analysis

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INTRODUCTION

- QR codes are widely used in daily life for payments, websites, and information sharing
- They provide quick access by simply scanning through mobile devices
- However, QR codes can hide malicious links without user awareness
- Attackers can use QR codes to redirect users to phishing or fake websites
- Users usually scan QR codes without verifying their safety
- This creates serious security risks like data theft and financial fraud
- This project focuses on detecting and analyzing QR code threats using Java



PROBLEM STATEMENT

- QR codes are widely used for payments, advertisements, websites, and information sharing.
- Users often scan QR codes without knowing the destination or security risks involved.
- Cybercriminals can use malicious QR codes to redirect users to phishing, fake payment, or malware websites.
- Traditional QR scanners only decode the content and do not analyze potential threats.
- There is a need for a system that can assess QR code safety before users access the embedded links.
- This project aims to provide risk-based QR code analysis and threat detection using Java.

OBJECTIVES

- To develop a Java-based system for analyzing QR codes and detecting security threats.
- To decode QR codes and extract embedded URLs from uploaded images.
- To identify suspicious URLs, phishing attempts, and potentially malicious links.
- To evaluate URL safety using multiple security parameters and generate a risk score.
- To provide clear, explainable results that help users make safer decisions before opening QR code links.



LITERATURE REVIEW

AUTHOR (YEAR)	PAPER TITLE	KEY FINDINGS & RELEVANCE
Singh & Kumar (2025)	QR Code Security: Emerging Threats and Countermeasures	Identified phishing attacks through malicious QR codes. Highlights the need for QR code security analysis before users open links.
Chen et al. (2024)	Detection of Malicious URLs Using Risk-Based Analysis	Demonstrated that URL features such as HTTPS status, suspicious keywords, and URL patterns are effective indicators of malicious intent.
NIST Cybersecurity Report (2024)	Guidelines for Phishing and Social Engineering Prevention	Emphasized phishing URLs as a major cybersecurity threat and recommended proactive URL verification mechanisms.
Ahmed & Rashid (2024)	Explainable Security Analysis for Web Threat Detection	Proposed explainable threat analysis instead of simple Safe/Unsafe classification, improving transparency and user trust.

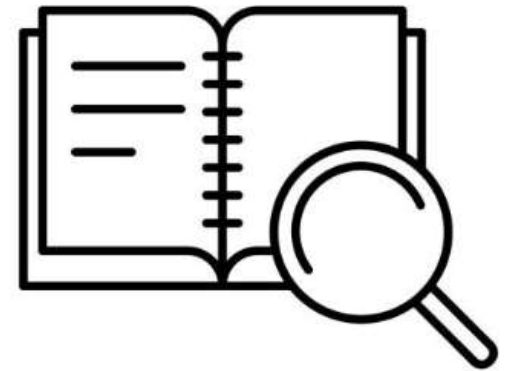
SYSTEM ARCHITECTURE

QR Code Security Analysis Process



SCHEDULING ALGORITHM

- The system follows a **sequential workflow model** for processing QR codes.
- QR code image is received and decoded to extract the embedded URL.
- The extracted URL is validated and analyzed for security threats.
- Multiple security checks are performed to identify suspicious patterns.
- A risk score is generated based on the detected threats.
- The final result is displayed with the risk score and threat explanation.




```

1- import java.net.URI;
2- import java.util.Arrays;
3- import java.util.List;
4
5- public class ThreatAnalyzerDemo {
6
7-     public static void main(String[] args) {
8
9         String rawValue = "http://192.168.1.1/login-payment.xyz";
10
11         // String Handling
12         String normalized = rawValue.trim().toLowerCase();
13
14         System.out.println("Original URL : " + rawValue);
15         System.out.println("Normalized URL : " + normalized);
16
17         // Exception Handling
18         try {
19
20             URI uri = URI.create(normalized);
21
22             String domain = uri.getHost();
23             String path = uri.getPath();
24
25             System.out.println("\nDomain : " + domain);
26             System.out.println("Path : " + path);
27
28             // String Split
29             String[] labels = domain.split("\\.");
30
31             System.out.println("\nDomain Parts:");
32
33             // Last Index & Substring
34             int index = domain.lastIndexOf('.');
35             String tld = domain.substring(index + 1);
36
37             System.out.println("\nTop Level Domain : " + tld);
38
39             // Keywords
40             List<String> keywords = Arrays.asList(
41                 "login",
42                 "verify",
43                 "payment",
44                 "upi",
45                 "wallet",
46                 "bank"
47             );
48
49             System.out.println("\nKeyword Detection:");
50
51             // Loop + contains()
52             for (String word : keywords) {
53
54                 if (normalized.contains(word)) {
55                     System.out.println(word + " Found");
56                 }
57             }
58
59             // Regex Matching
60             if (normalized.matches("(?i)^https?://.+")) {
61                 System.out.println("\nValid URL");
62             }
63
64             // IP Address Validation
65
66             String ip = "192.168.1.1";
67
68             String[] raw = ip.split("\\.");
69
70             int[] p = new int[4];
71
72             // For Loop
73
74             for (int i = 0; i < 4; i++) {
75                 p[i] = Integer.parseInt(raw[i]);
76             }
77
78             boolean valid = true;
79
80             // Enhanced For Loop
81             for (int n : p) {
82
83                 if (n < 0 || n > 255) {
84                     valid = false;
85                 }
86             }
87
88             System.out.println("\nValid IP : " + valid);
89
90             // Path Analysis
91             System.out.println("\nPath Segments:");
92             for (String segment : path.split("/")) {
93
94                 if (!segment.isBlank()) {
95                     System.out.println(segment);
96
97                     String clean = segment.toLowerCase();
98
99                     int vowels = clean.replaceAll("[^aeiou]", "").length();
100                     int digits = clean.replaceAll("[^0-9]", "").length();
101
102                     System.out.println("Vowels : " + vowels);
103                     System.out.println("Digits : " + digits);
104                 }
105             }
106
107             catch (NumberFormatException e) {
108                 System.out.println("Invalid IP Address");
109             }
110
111             // Exception Handling
112             catch (Exception e) {
113                 System.out.println("Invalid URL Format");
114             }
115         }
116     }
117 }

```


OUTPUT

QR Threat Analyzer
SCAN BEFORE YOU TRUST

HomeUpload QRDashboardVisualizationReportsAboutScan Now

Upload QR Image

Drag & drop, browse, or try a sample. We inspect the destination — we never tap the link.

✔ QR Detection Ready

✔ Image Analysis Ready

✔ Threat Analysis Ready



Drop your QR code here

PNG · JPG · JPEG — up to 10MB

Choose File

Accepted Formats

PNG · JPG · JPEG

Try a sample

Suspicious payment link

Trusted destination

Raw IP phishing

Privacy

QR decoding and threat analysis happen in your browser. The destination link is inspected, never visited or executed.

Smart QR Code Threat Analyzer

Smart QR Threat Assessment Report · Scan Before You Trust

VERDICT

SAFE RISK

6%



Decoded URL

<https://stego-threat.assut.edu.eg/>

Final Domain

assut.edu.eg

Domain Age

Not verified

Findings

11

URL Shortener Detected

No URL shortening service found.

11

Domain Age < 30 Days

Domain registration age is not verified for this report.

10

Domain Age Not Verified

A domain with unknown registration age has been associated with a verified long-term domain.

11

Suspicious Keyword Found

No high-risk keywords found.

11

Multiple Redactions

Search or long-link obscuration.

11

Users Raw IP Address

Resolved to a valid domain.

11

Private/Internal IP Address

Decoded domain is a public address.

11

Suspicious Top-Level Domain

Domain uses a nonstandard TLD.

11

Hidden Destination / QR Redirector

The scanned URL points directly to its visible destination.

11

Opaque URL Token

The URL path is neither readable or empty.

11

Deceptive Host Structure

Redundant characters in domain.

11

Insecure Connection (No HTTPS)

Final destination is served over HTTP(S).

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CONCLUSION

- The Smart QR Code Threat Analyzer successfully detects and analyzes QR code risks
- It helps users identify malicious or suspicious QR codes before opening links
- The system decodes QR codes and extracts embedded URLs efficiently
- It reduces the chances of phishing attacks and online fraud through QR codes
- The Java-based implementation ensures simplicity and reliability
- Overall, the system improves user safety in QR code usage and digital transactions



REFERENCES

Java SE Documentation – Oracle

ZXing (“Zebra Crossing”) QR Code Library Documentation

IEEE Papers on QR Code Security and Phishing Detection

Research articles on URL-based threat analysis and cyber security systems

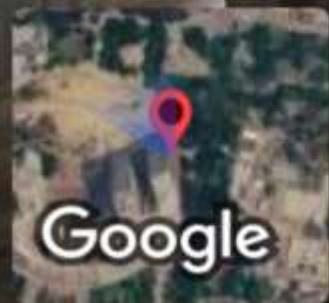
Cybersecurity guidelines related to phishing and malicious link detection

Online Java tutorials and documentation for networking and file handling

Standard textbooks on Object-Oriented Programming using Java

Articles on real-world QR code misuse in digital payment systems

THANK YOU



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